YEAR 11 MATHS METHODS INVESTIGATIVE TASK 4

Introduction to differential calculus

Prerequisites

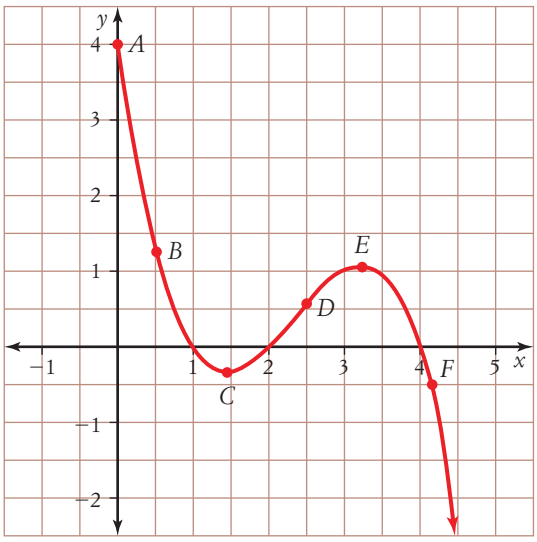
- Sketching the graphs of polynomials
- Understanding rates of change and speed as a rate of change, including limits and finding the derivative from first principles
- Understanding the relationship between a function and its gradient (derivative)
- Understanding the application of calculus to find the maximum and minimum values of polynomials

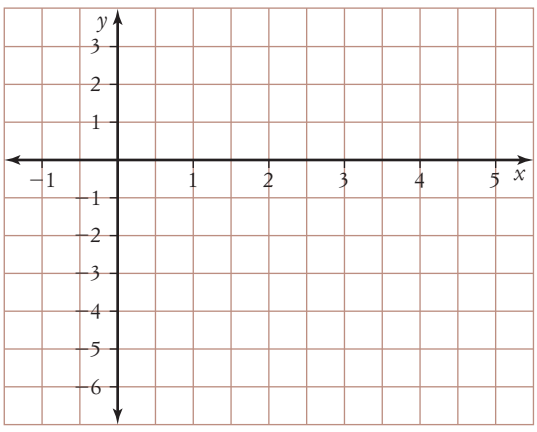
Students will need a printout of the diagrams in question **1**.
Graph paper will be useful.
Calculators of any kind may be used, but marks will be deducted if working out is asked for and not shown.

[Total: 80 marks]

1 a Consider the function $y = f(x)$ in the top diagram below. Draw a tangent at the indicated points to estimate the gradient at each of the points A to F.

b Plot the gradient against the corresponding x value to draw the gradient function $y = f'(x)$.





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c Explain how to use the graph of $y = f'(x)$ to decide if a point where $f'(x) = 0$ represents a local maximum, a local minimum or an inflection point of the function $y = f(x)$. Use your answers to confirm the nature of all points where $f'(x) = 0$ in the graphs above.

d Given that $g(x) = (x - 1)(x - 2)(x - 4)$, find an expression for $g'(x)$ from first principles.

e Use the x- and y-intercepts to determine an equation for $f(x)$, given it is a cubic.

f Express $f(x)$ in terms of $g(x)$, and hence deduce the derivative of $f(x)$. Use this result to check your result for the gradient at points A and D. *Note*: Marks will not be deducted if the answers are not exactly the same.

[6 + 4 + 6 + 5 + 3 + 4 = 28 marks]

2 Consider the graph of $y = x^4 - 2x^3 + 1$.

a Write down the derivative of $x^4 - 2x^3 + 1$.

b Find and describe all stationary points on the curve.

c How does the curve behave as $x \longrightarrow \infty$ and as $x \longrightarrow -\infty$?

d Use your calculator to find any other significant points and sketch the curve.

e Find the equation of the tangent to the curve at the point where $x = 2$. All working should be shown.

f Draw in the tangent on your sketch in part **d**.

[2 + 7 + 3 + 6 + 4 + 2 = 24 marks]

3 Two positive numbers add to ten. Find the maximum product of the two numbers:

a using a table

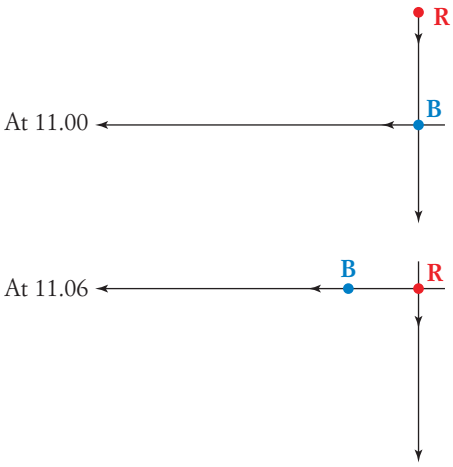
b using a graph (A CAS calculator may be used.)

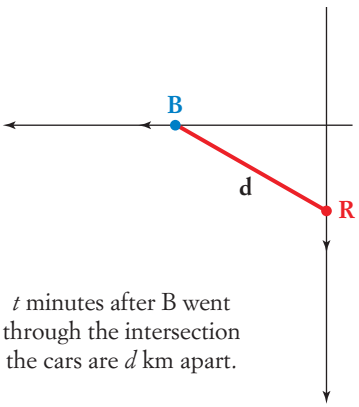
c using calculus.

[2 + 2 + 6 = 10 marks]

4 Two cars are approaching an intersection. The blue car is travelling west at 30 km/h. The red car is travelling south at 60 km/h.

The blue car went through the intersection at 11.00 a.m. The red car travelled through the intersection six minutes later. Both had a green light and both continued at a constant velocity.





t minutes after B went through the intersection the cars are d km apart.

a i Convert both speeds to km/min.

ii How far has the blue car travelled t minutes after 11.00 a.m.? Find an expression for the blue car's position t minutes after 11.00 a.m.

iii How far has the red car travelled past the intersection t minutes after 11.00 a.m.?

iv Find an expression for the distance d between the two cars t minutes after 11.00 a.m.

b At what time will the cars be closest together?

c Find their distance apart at this time in the clearest exact form as well as in an approximate form.

[2 + 2 + 5 + 2 + 3 + 4 = 18 marks]

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